Detección de Anomalías Acústicas en Maquinaria Industrial

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Problema

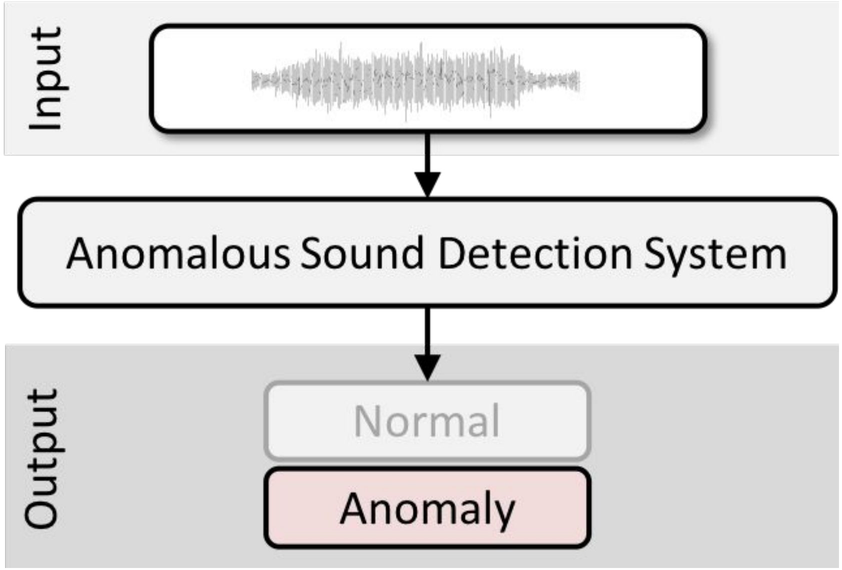
Problema

- Tiempos de inactividad no planificados

Problema

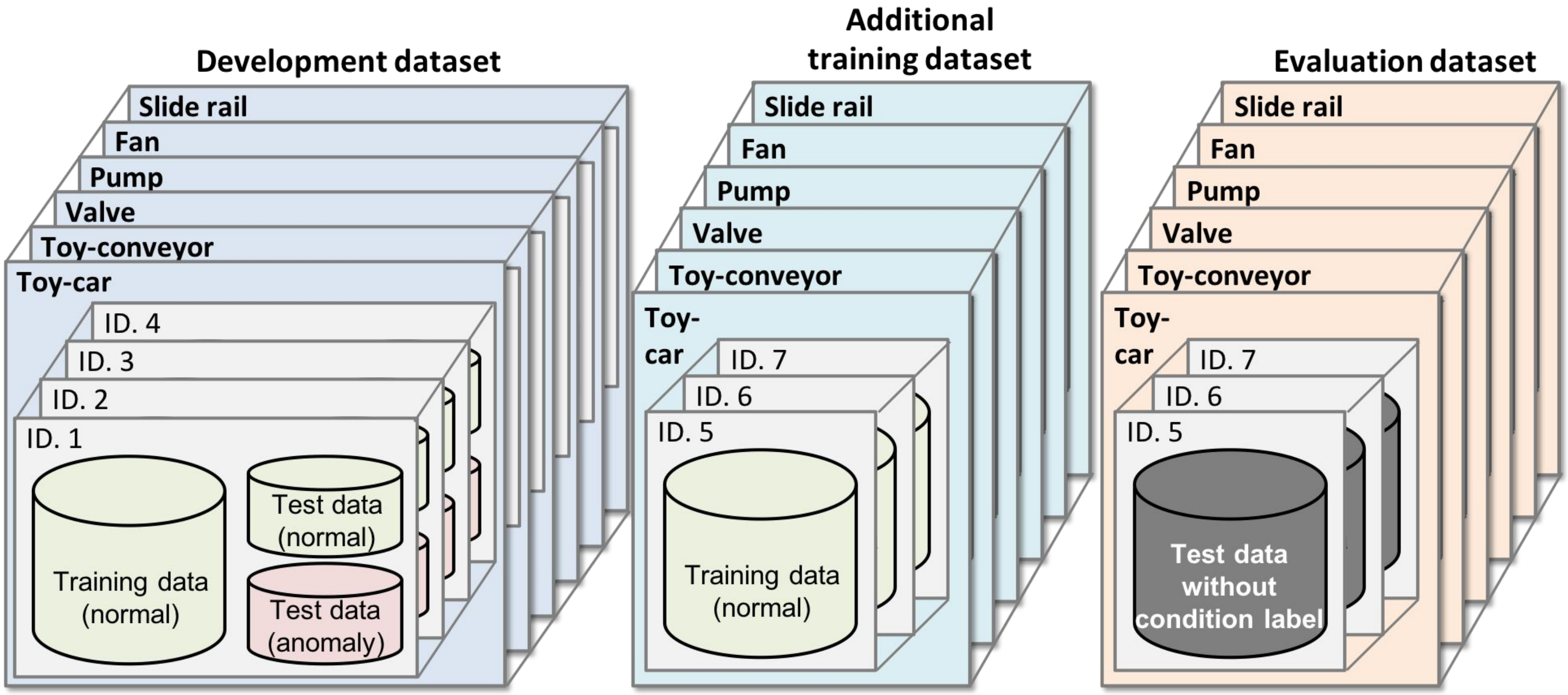
- Tiempos de inactividad no planificados
- Mantenimiento reactivo

Alcance

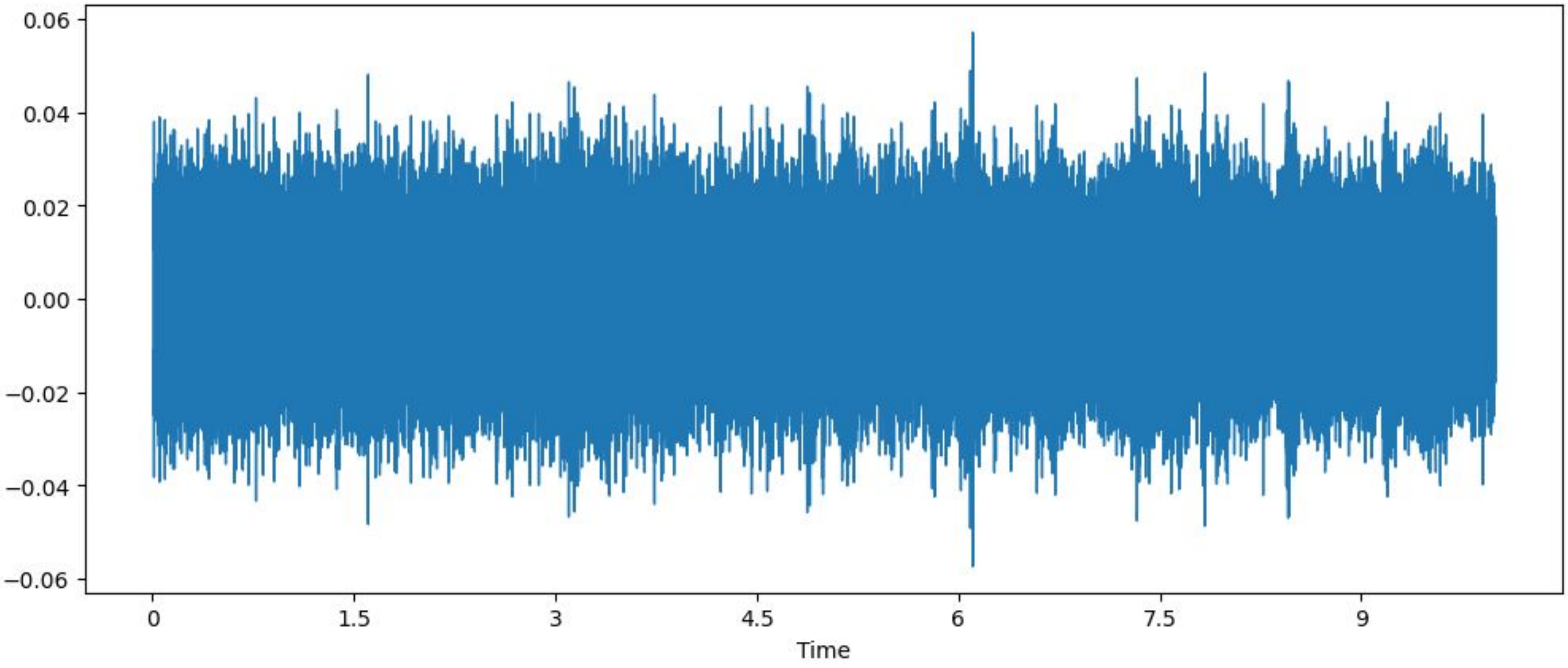


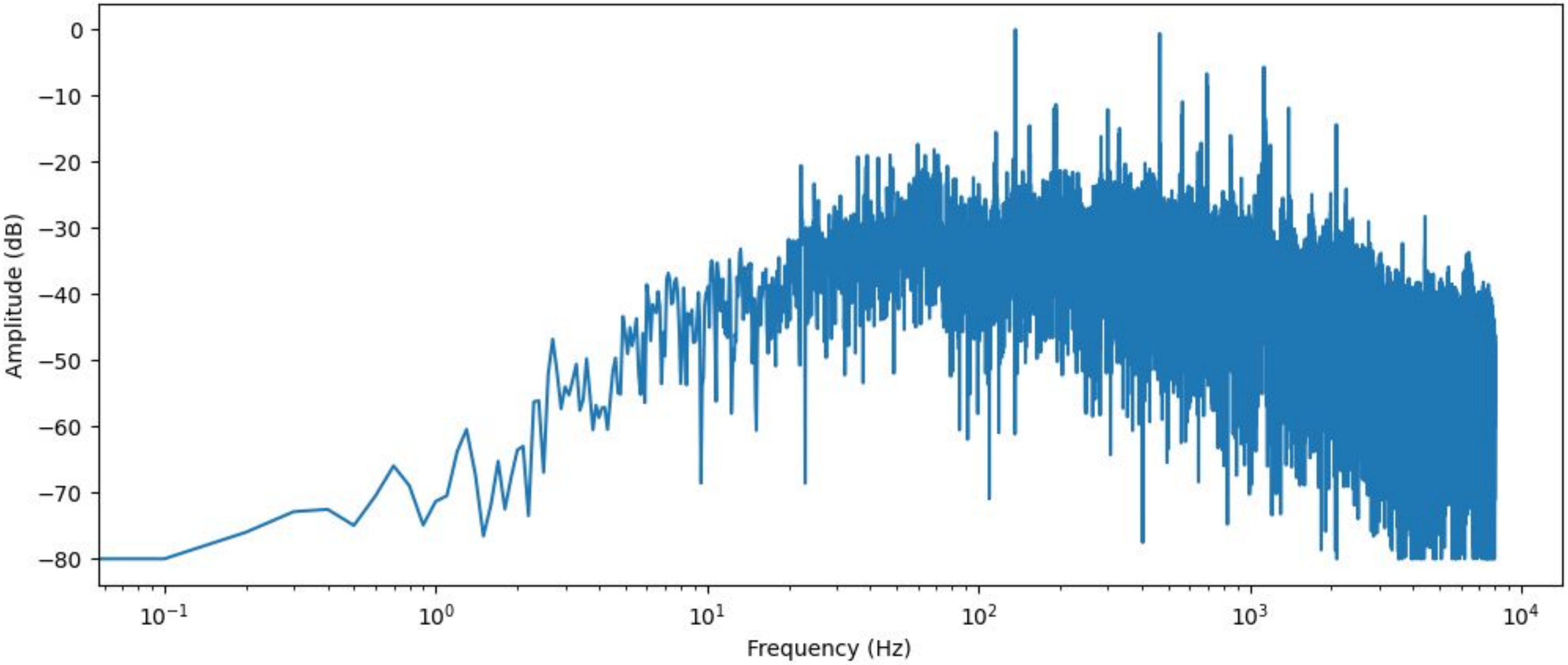
Detectar anomalías acústicas en maquinaria industrial mediante el análisis acústico, haciendo uso de autoencoders

Dataset

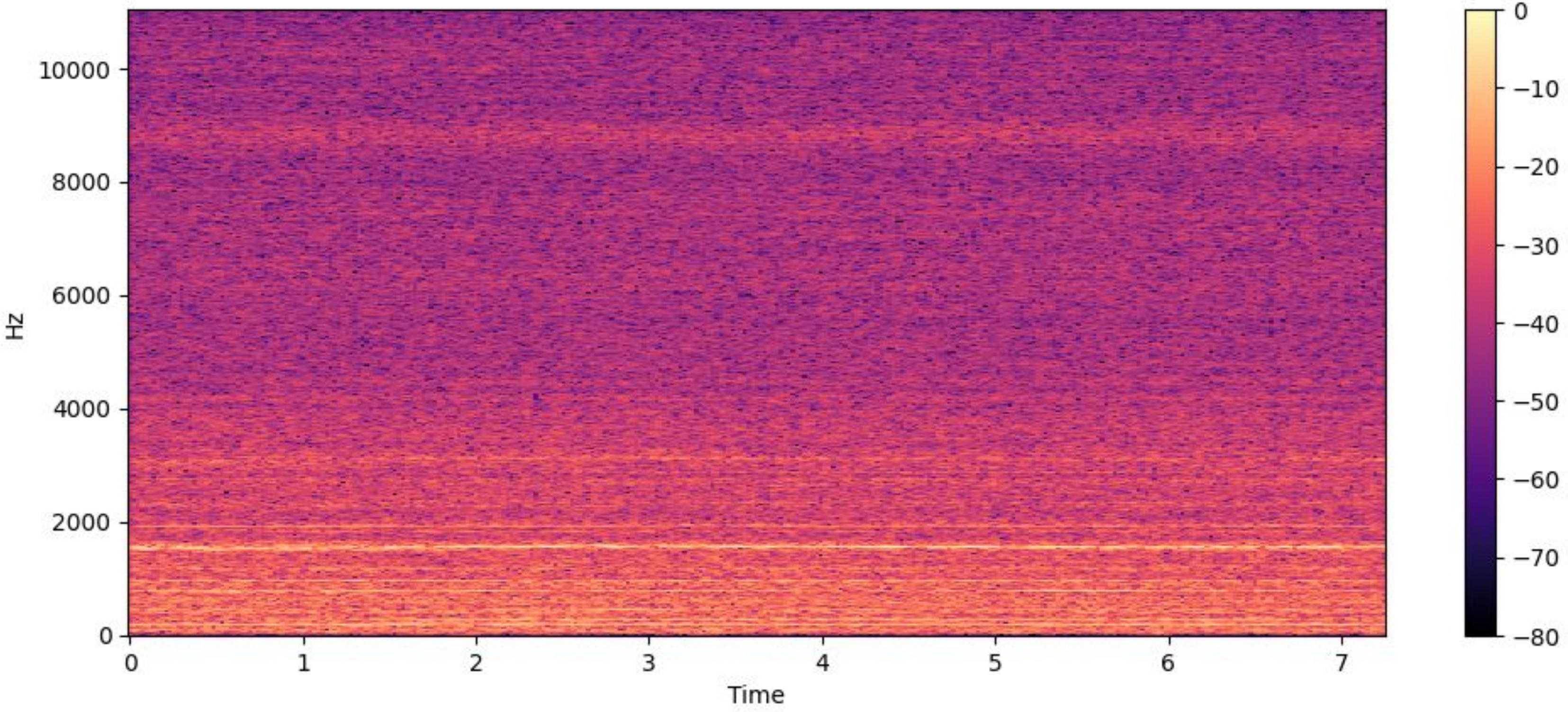


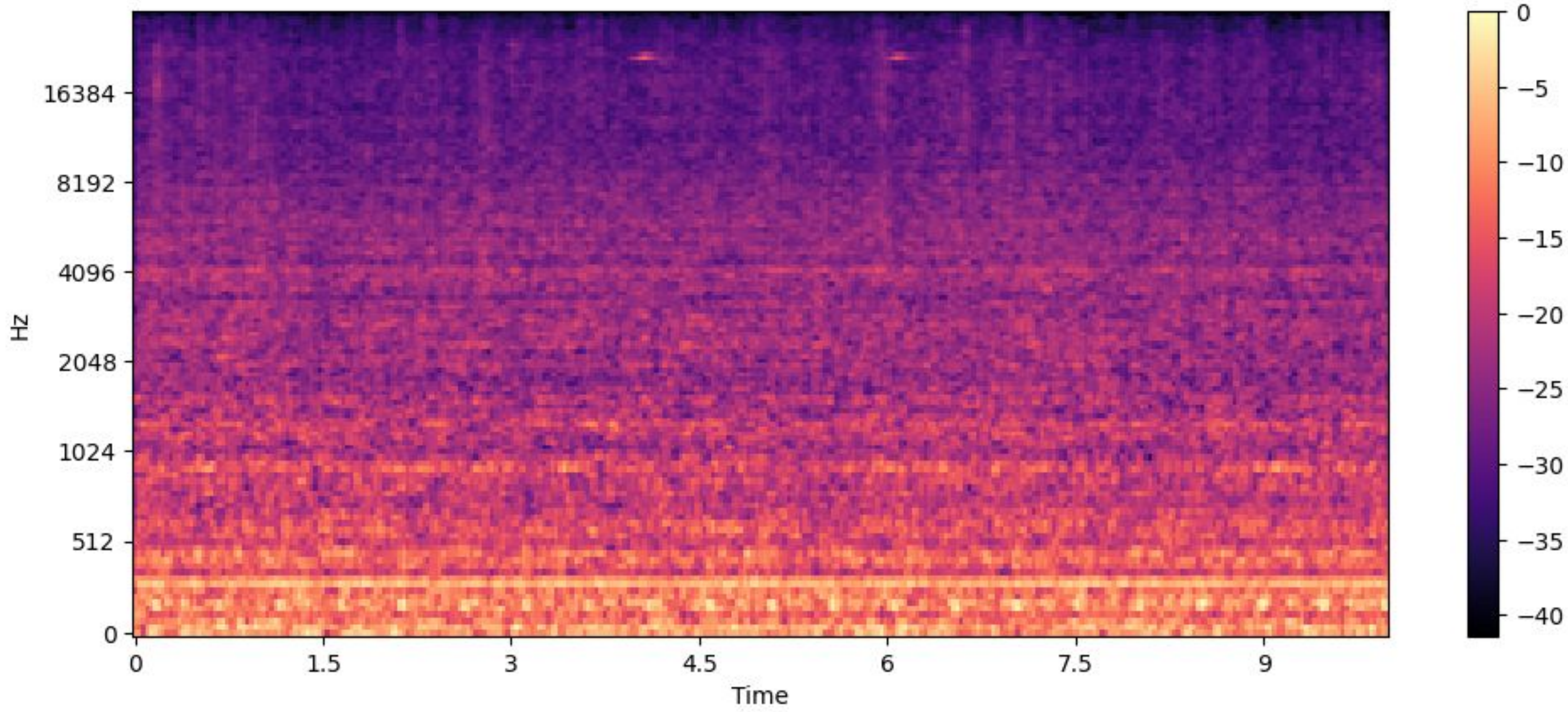
Development dataset
Slide rail
Fan
Pump
Valve
Toy-conveyor
Toy-car
ID. 4
ID. 3
ID. 2
ID. 1
Training data (normal)
Test data (normal)
Test data (anomaly)
Additional training dataset
Slide rail
Fan
Pump
Valve
Toy-conveyor
Toy-car
ID. 7
ID. 6
ID. 5
Training data (normal)
Evaluation dataset
Slide rail
Fan
Pump
Valve
Toy-conveyor
Toy-car
ID. 7
ID. 6
ID. 5
Test data without condition label


0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
0
1.5
3
4.5
6
7.5
9
Time

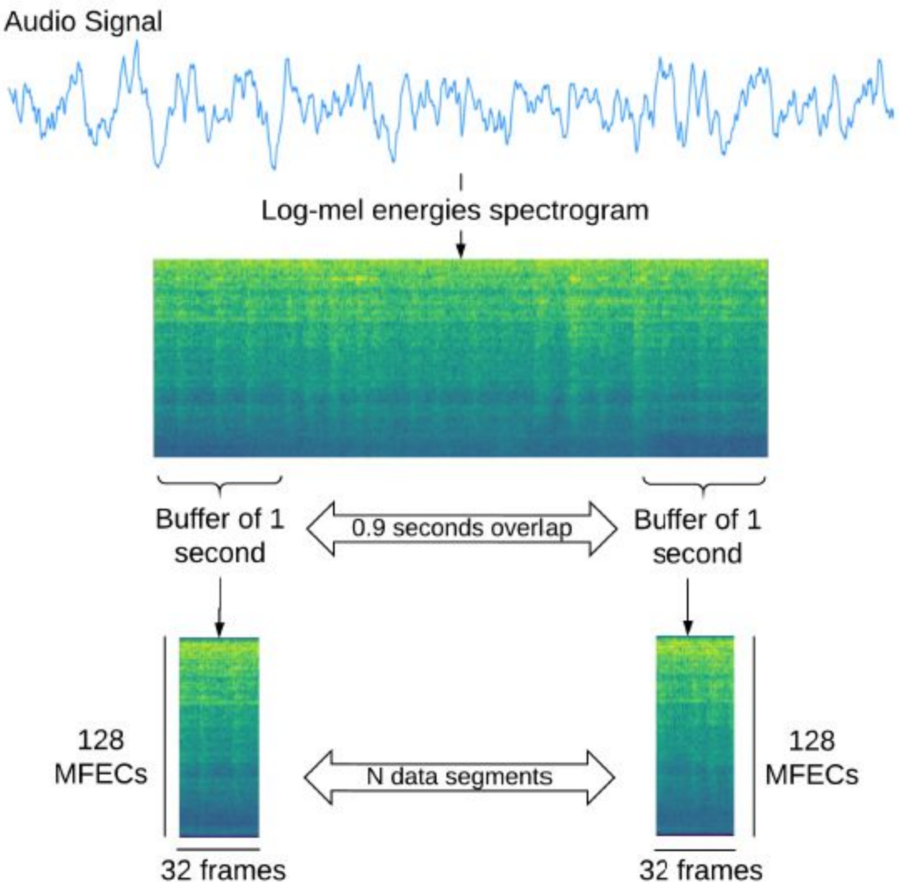


0
−10
−20
−30
−40
−50
−60
−70
−80
Amplitude (dB)
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
Frequency (Hz)


Hz
10000
8000
6000
4000
2000
0
0
1
2
3
4
5
6
7
Time
0
−10
−20
−30
−40
−50
−60
−70
−80



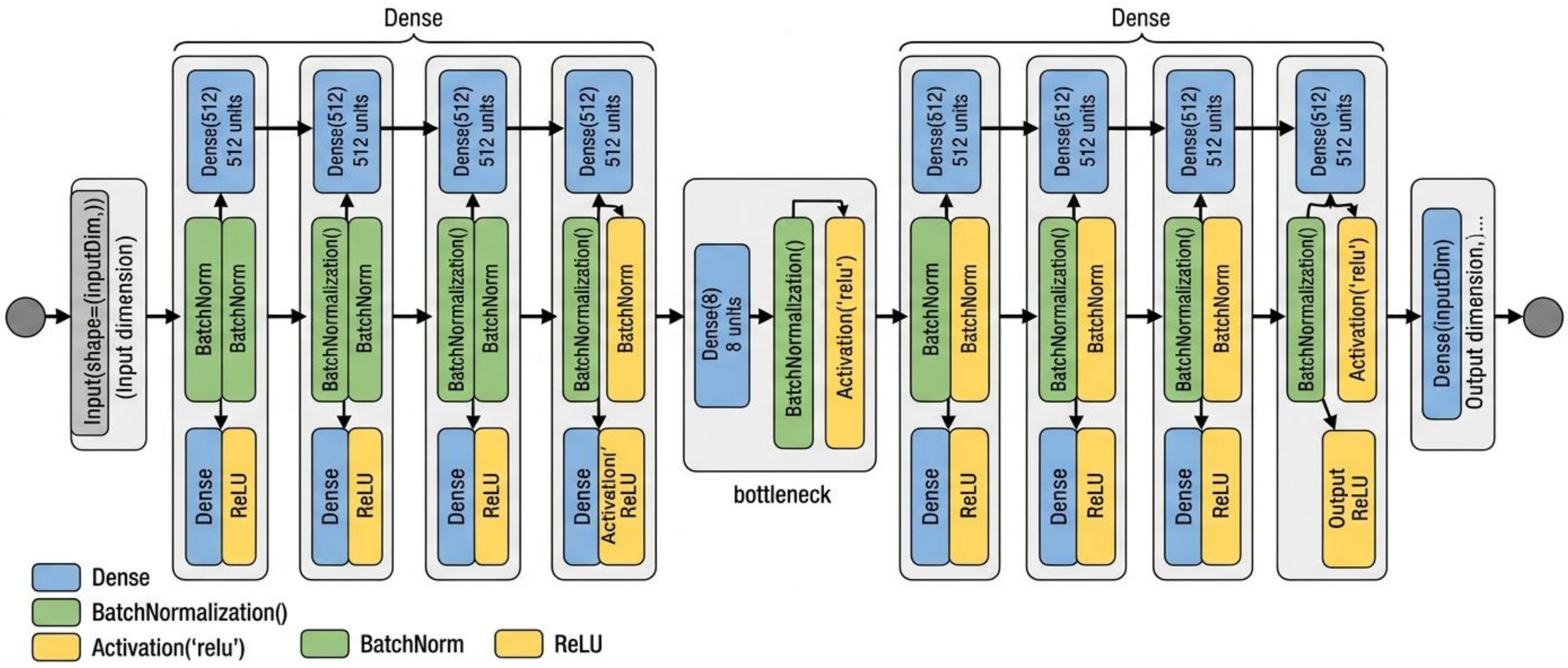
Hz
16384
8192
4096
2048
1024
512
0
0
1.5
3
4.5
6
7.5
9
Time
0
−5
−10
−15
−20
−25
−30
−35
−40


Audio Signal
Log-mel energies spectrogram
Buffer of 1 second
0.9 seconds overlap
Buffer of 1 second
128 MFECs
N data segments
128 MFECs
32 frames
32 frames

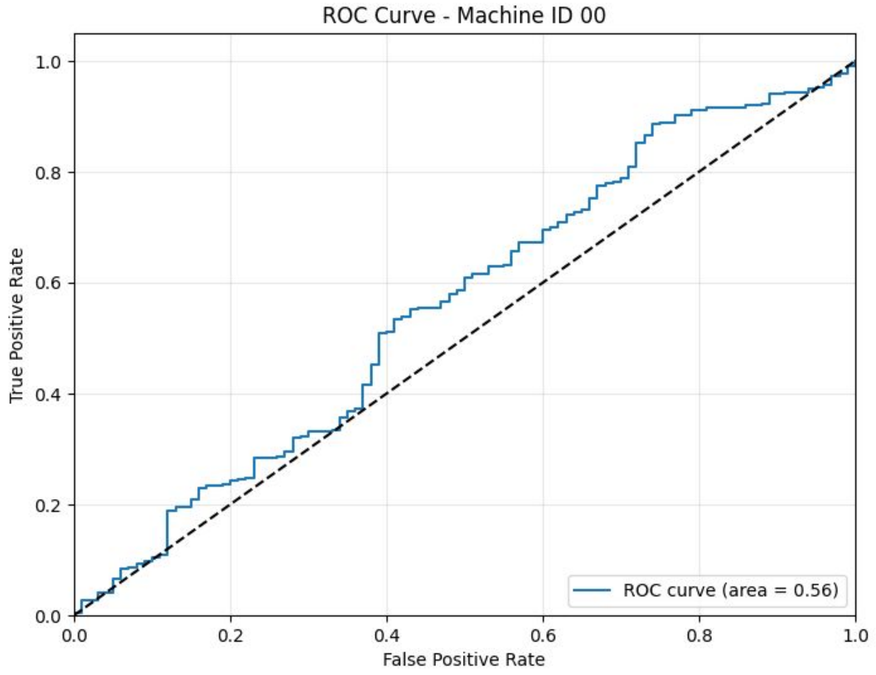
Muestras de Audio

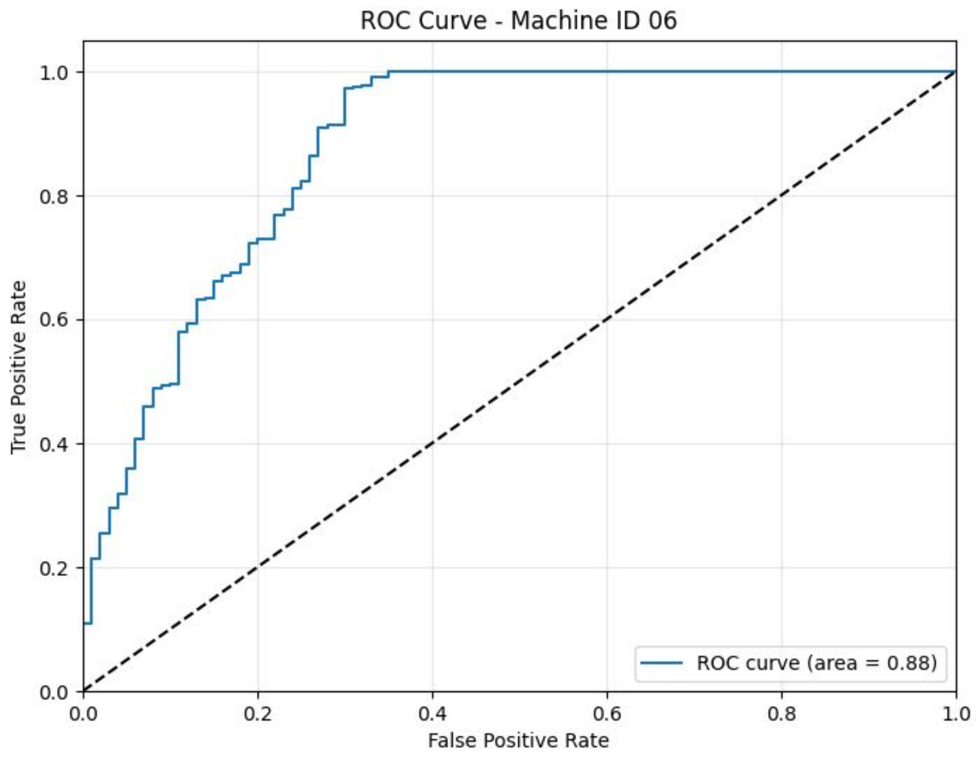
Arquitectura de Red Neuronal Profunda Autoencoder

```
def get_model(inputDim):
    return = get_model(inputDim), + recor(...)
```



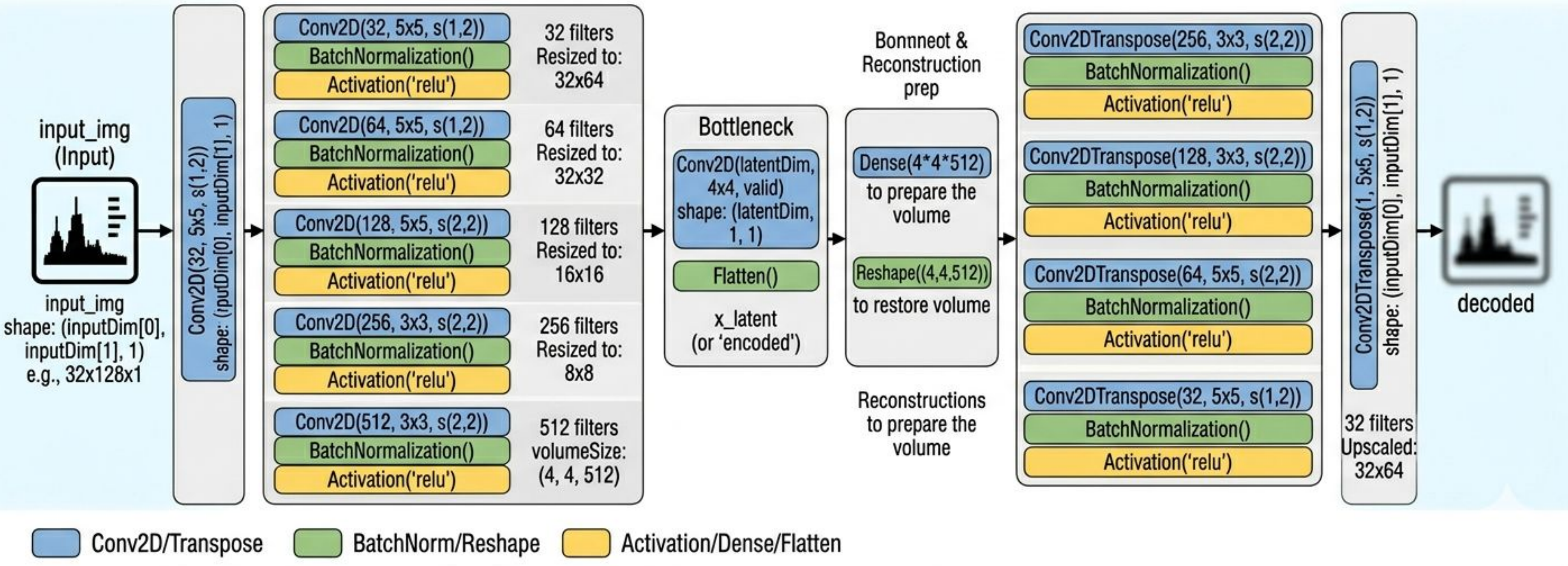
Machine ID	AUC
00	0.559091
02	0.812006
04	0.632299
06	0.879003



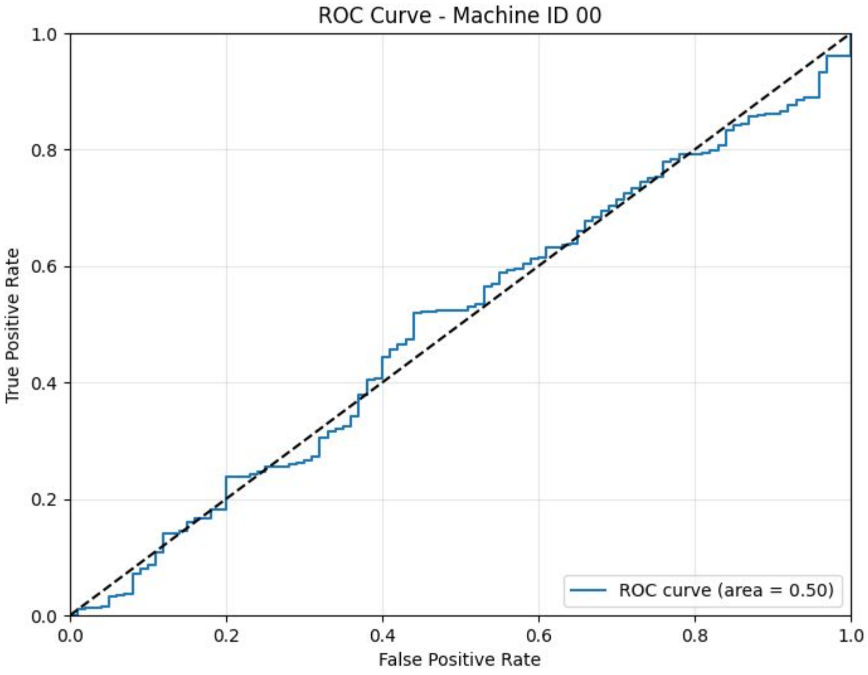

ROC Curve - Machine ID 06
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC curve (area = 0.88)

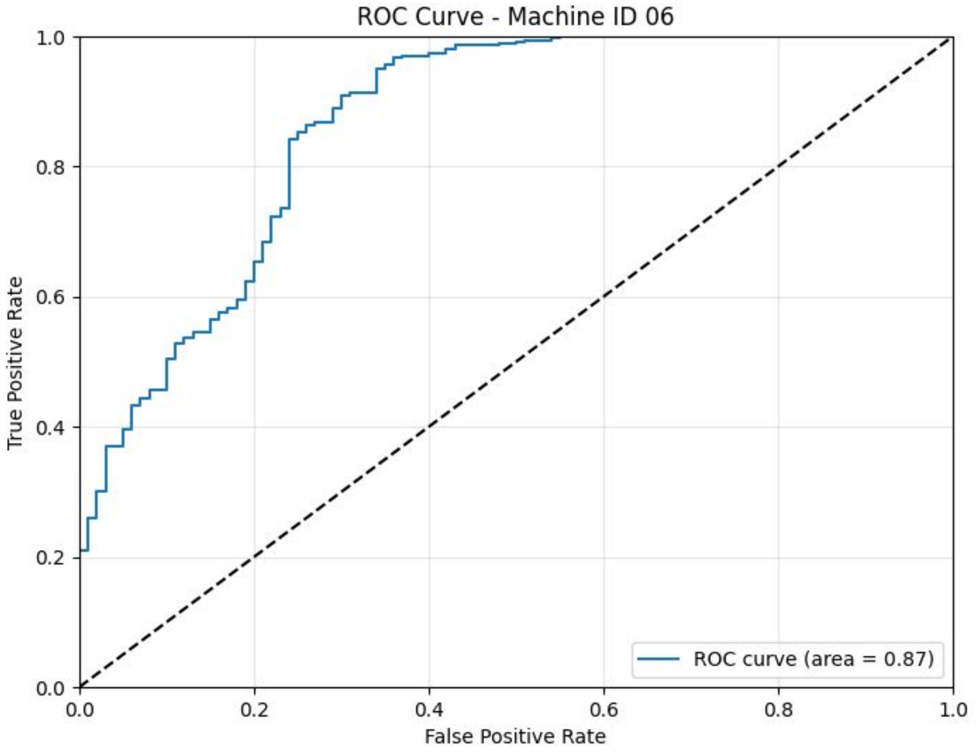
Arquitectura de Red Neuronal Profunda Autoencoder Convolucional Simple (Imagen/Espectrograma 2D)

```python
def get_model(inputDim, latentDim):
    input_img = Input(shape=(inputDim[0], inputDim[1], 1))

    # 5 Convolutional Encoder blocks with reduction strides
    encoded = ... (see below)

    # Bottleneck and Flattening
    x_latent = ...

    # 4 Deconvolutional (Conv2DTranspose) Decoder blocks
    decoded = ... (see below)

    return Model(inputs=input_img, outputs=decoded)
```



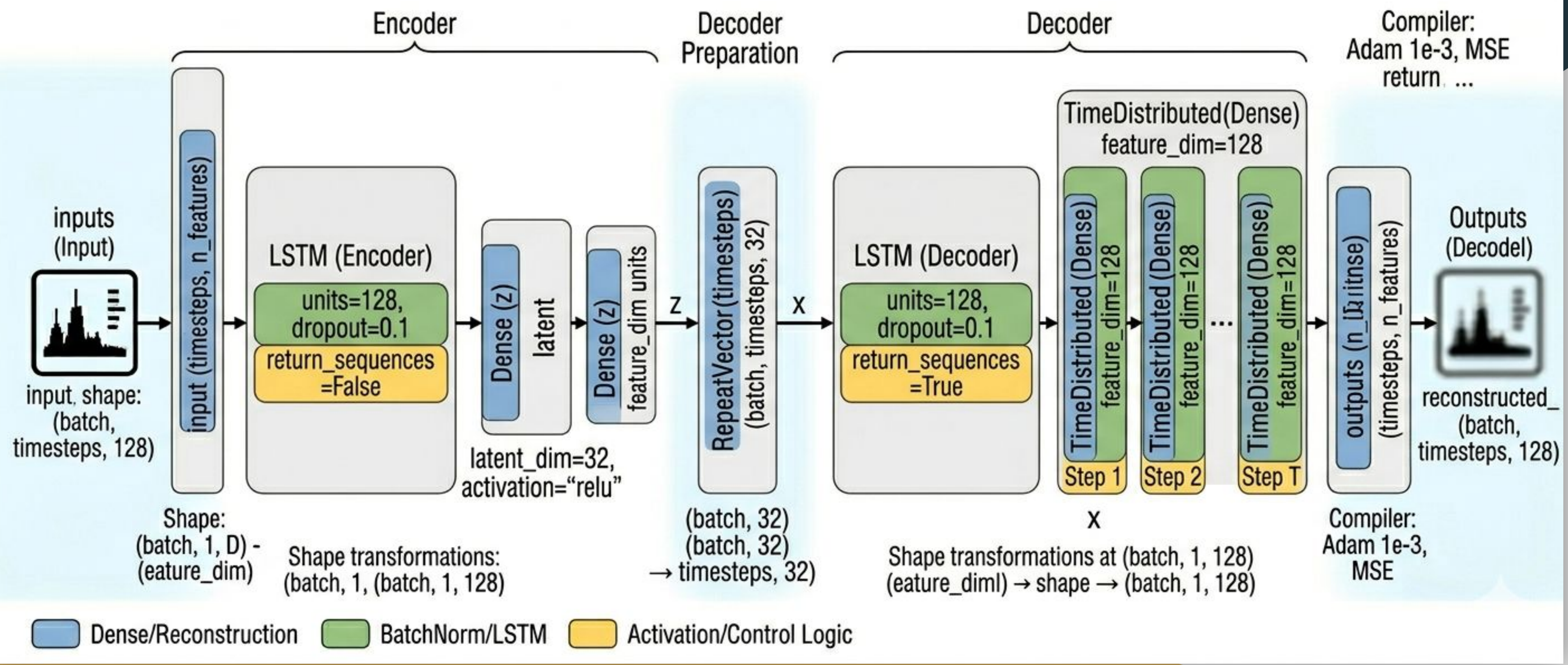
Machine ID	AUC
00	0.498182
02	0.775933
04	0.659167
06	0.865291



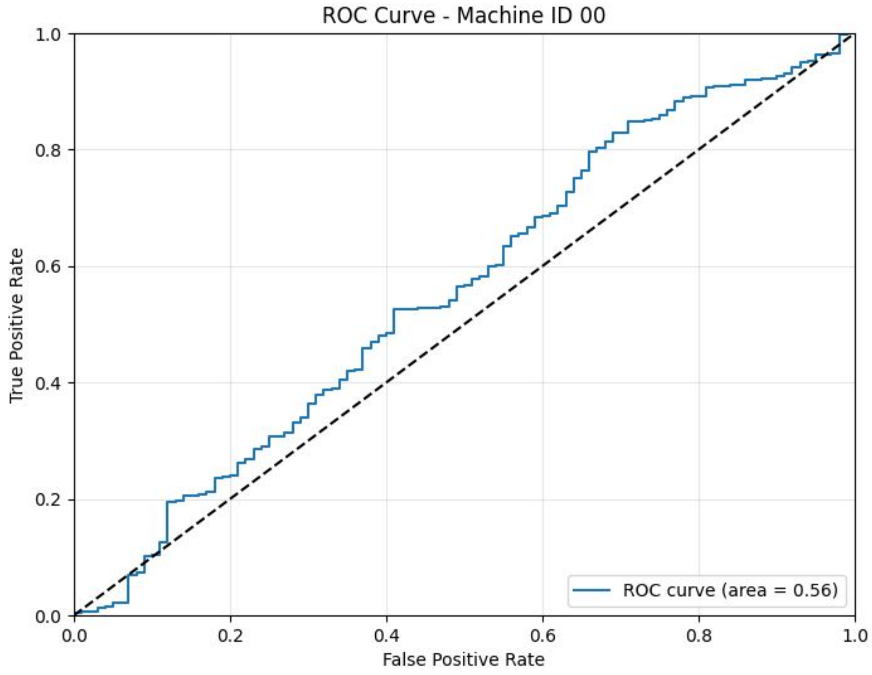

ROC Curve - Machine ID 06
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC curve (area = 0.87)

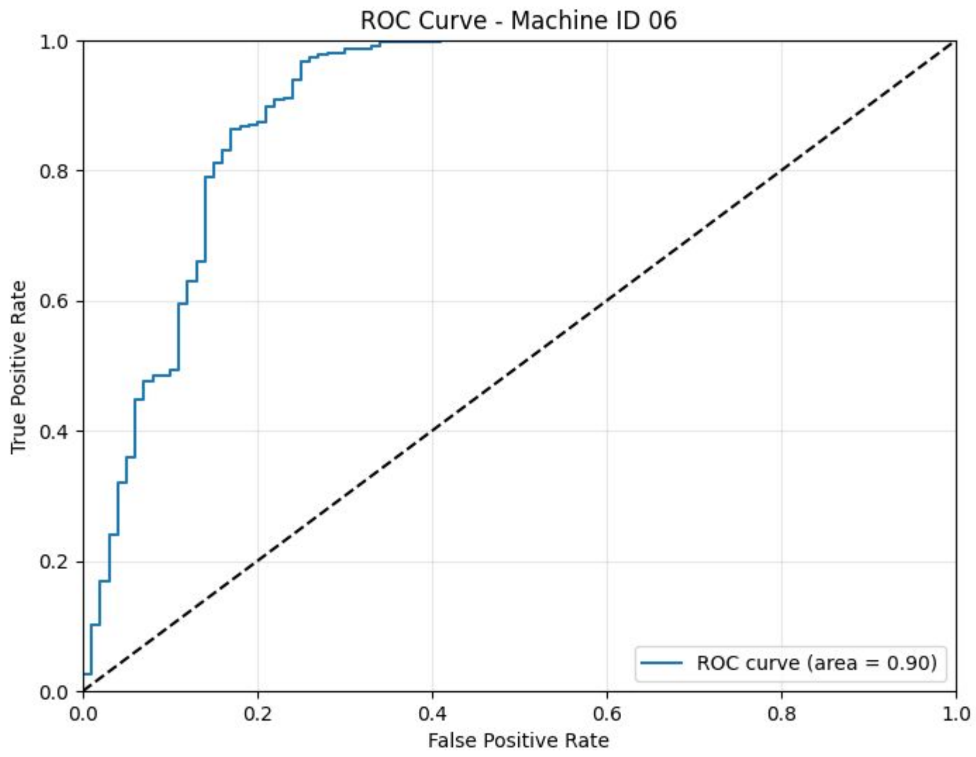
Seq2Seq LSTM Autoencoder Network Architecture (Vector-as-Sequence)

```
def build_seq2seq_lstm_ae(timesteps: int, n_features: int = 128, latent_dim: int = 32, units: int = 128, dropout: float = 0.1):
    ...
    return Model(timesteps, n_features, latent_dim))            # compiler(input_img, t_img)
```



Machine ID	AUC
00	0.557789
02	0.744930
04	0.560172
06	0.898670




ROC Curve - Machine ID 06
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC curve (area = 0.90)

Entrega Final

- Uso del dataset completo
- U-Net Autoencoder
- Variational Autoencoder

Conclusiones

- La obtención de los datos debe ser lo mejor posible
- Los autoencoders son apropiados para detección de anomalías acústicas

Notebook